INDUS UNIVERSITY

Monthly Progress Report - Software Group Project

(Report: 5)

Academic Year: 2026-27 Month: July

Project Title: House price prediction using AI/ML

Project Domain: AI/ML

Group Number: 50 Project Guide: Dhwani Mam

Student Details:

Enrolment No.	Student Name	Contribution	Signature
IU2341050103	Ravi Patel	➢ Flask Web Application Support ➢ HTML Interface Development ➢ User Input Form Design ➢ UI Testing	
IU2341050105	Kenil Patel	➢ Model Deployment Support ➢ Prediction Result Integration ➢ Application Testing ➢ Error Handling Support	
IU2341050114	Vishnu Jikadara	➢ Flask Backend Development ➢ ANN Model Integration ➢ Input Preprocessing & Feature Encoding ➢ Real-Time Prediction Functionality	

Project Objective:

To integrate the trained Artificial Neural Network (ANN) model into a Flask-based web application, develop a user-friendly interface for entering property details, perform real-time house price prediction, and test the complete prediction system.

Monthly Work Completed

• Developed a Flask-based web application for the House Price Prediction system.

• Integrated the trained ANN model (ann_model.keras) into the Flask application.

• Loaded the saved StandardScaler used during model training.

• Loaded the saved feature column information for consistent input feature generation.

• Implemented user input handling using Flask GET and POST requests.

• Created a 189-feature input vector according to the trained ANN model structure.

• Implemented one-hot encoding for Area Type and Location features.

• Implemented handling of locations that are not directly available in the trained feature columns.

• Applied the previously trained StandardScaler to the user input before prediction.

• Integrated ANN prediction functionality into the Flask backend.

• Developed the HTML-based user interface for entering property details.

• Added input fields for Total Sqft, Bathroom, Balcony, BHK, Area Type, and Location.

• Added searchable Location selection functionality.

• Added prediction result display on the web interface.

• Added validation handling for invalid property details.

• Designed and applied CSS styling to improve the basic user interface.

• Tested the complete Flask application locally.

• Performed functional testing using multiple property configurations.

Functional Testing Results and Screenshots

House Price Prediction

Total Sqft:

1200

Bathroom:

2

Balcony:

1

BHK:

2

Area Type:

Super built-up Area

Location:

Whitefield

Predict Price

Predicted Price: ₹68.92 Lakhs

House Price Prediction

Total Sqft:

1500

Bathroom:

3

Balcony:

2

BHK:

3

Area Type:

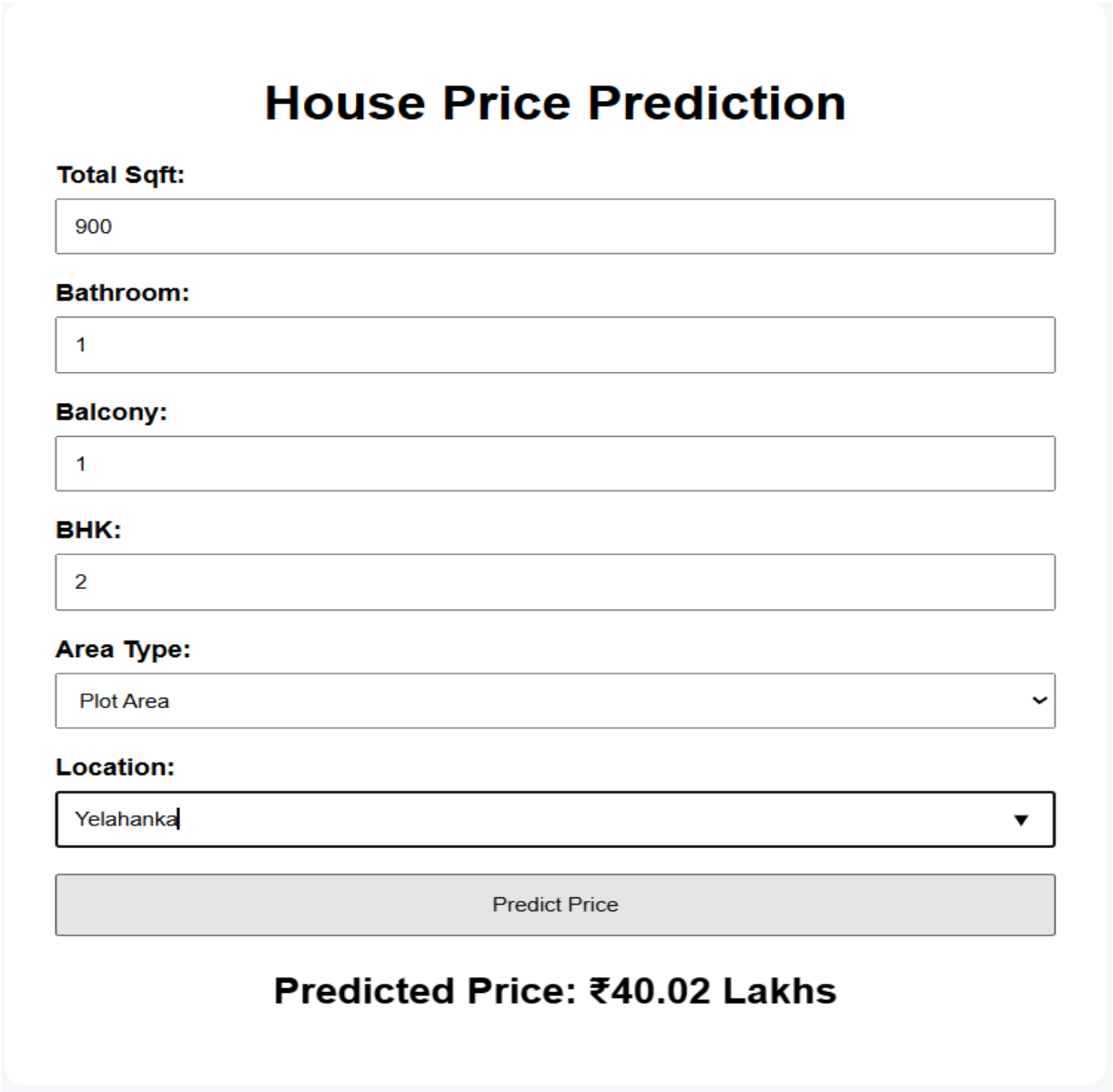
Built-up Area

Location:

Electronic City

Predict Price

Predicted Price: ₹56.16 Lakhs



Test	Property Details	Area Type	Location	Predicted Price
1	1200 sqft, 2 BHK, 2 Bath, 1 Balcony	Super built-up Area	Whitefield	₹68.92 Lakhs
2	1500 sqft, 3 BHK, 3 Bath, 2 Balcony	Built-up Area	Electronic City	₹50.21 Lakhs
3	900 sqft, 2 BHK, 1 Bath, 1 Balcony	Plot Area	Yelahanka	₹40.02 Lakhs

Challenges Faced

- Integrating the trained ANN model with the Flask application.
- Maintaining the same feature structure during training and prediction.
- Handling one-hot encoded Area Type and Location features.
- Handling differences in spacing within categorical feature names.
- Applying the previously fitted scaler correctly to new user inputs.
- Connecting HTML form inputs with Flask backend processing.
- Testing and debugging the complete end-to-end prediction workflow.
- Designing a simple and user-friendly interface for real-time prediction.

Next Month Plan

- Further improve the user interface and overall website design.
- Perform more extensive functional and edge-case testing.
- Improve input validation and error handling.
- Complete final integration of all project components.
- Prepare project documentation and final screenshots.
- Deploy the complete House Price Prediction System.
- Perform final system testing and prepare the project for demonstration.

Guide Remarks

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Overall Progress :75 %

Student Signatures:

- Ravi Patel: _____
- Kenil Patel: _____
- Vishnu Jikadara: _____

Project Guide: _____